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Brainpower Education- Service to the human race

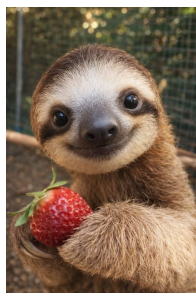
Teacher Code : HEWP **GOWS** **KERJ** **LEAB** SRIK FIRR **BUYR** **SHAA**
CHEW

Mathematical Methods 3/4

APPLICATION TASK 2

Question and Answer Book

The Adventures of the SAC-MAN 2

 dx

The elementary Integral

Reading time is **15 Minutes**

Writing time is **120 Minutes**

Approved Materials:

- Any aids for the sketching of graphs
- A physical **OR** digital Mathematica Bound Reference
- **OR** A CAS Calculator with **physical** bound reference
- Stationary including pencils and a scientific calculator (if desired)

Materials supplied:

- Question and Answer book of 19 pages



Contains

page

In Part I , The School of JKSS was at a crisis and quandary: SAC I was stolen by the Schmiddler and his gang of goons. Will's teacher Mr. Kermond, organized a search party to procure the missing SAC I , they found it and a special Chemical X resulted in William having strange powers , he is now the superhero "The SAC-MAN" , days later, a new initiative has began at the school , organized by an external party "The Windsweep Consortium". The initiative is named:

THE ELEMENTARY INTEGRAL

The motives of this initiative are really difficult to identify and Mr. Kermond is suspicious that it might be Schmiddler's attempt to wreak havoc once again on the JKSS regions .

Schmiddler cackling in his lair , realized that SAC I was too trivial for the cohort , and decided to make SAC 2 hell instead, Will and Kermond had mixed feelings on the subject



SECTION A:

- Answer **all** questions in the booklet
- All written responses need to be in English
- If not specified then provide **exact** values for each question
- Mathematica /CAS Syntax is **not** suitable as working out
- Show all mathematical reasoning if a question is of multiple marks
- The sections in italics are story only and not required for the assessment

Question 1 (12 Marks)

TIME : 8:40 AM (GMT+10)

MEASUREMENT UNITS: Meters

The Elementary Antiderivatives' worker *Gupta Gupta* has proposed a new way to introduce integration to William's Methods Class, *Mr. Kermond* observes with intent and is alarmed when they decide to whip out pseudocode for certain approximations .

William's classroom whiteboard can be modelled by a parabolic arch that is concave down . Let the classroom wall be placed on a cartesian plane, where the ground represents the x axis and the y axis is a line perpendicular to it. The leftmost intersection with the ground is located at the origin , and 6 metres east exists another intersection with the ground. The maximum height the board reaches is 50 metres upward from the ground .

a.

i. State the equation of the board's perimeter in the form $y = f(x)$ (2 Marks)

For the first problem *Gupta Gupta* asks William's friend *Roshan* to approximate the area bound by the graph of the perimeter with 3 trapeziums of equal width , shown is her working

$$A = \frac{h}{2} [f(0) + 2f(2) + 2f(4) + f(6)]$$

$$A = \frac{1}{2} [f(0) + 2f(2) + 2f(4) + f(6)]$$

$$A = \frac{1}{2} [0 + 400/10 + 400/8 + 0]$$

ii. *Roshan* has made n amount of errors, state these errors , the value of n^2 and state the actual approximation , correct to 2 decimal places (2 Marks)



In order to prove to Mr. Kermond that the trapezium method is accurate enough , William Patel wonders to what extent of accuracy it requires

Let k be the minimum amount of trapeziums required such that the approximation is under 6.7% error with the actual bounded area , let h be the minimum step size for this to be achieved .

b. Determine the value(s) of $\log_h(k)$ correct to 2 decimal places (3

Marks)

Suddenly out of nowhere, a beam suddenly falls and nearly hits *Gupta Gupta* , the beam passes through the point $(0,p)$ and intersects tangentially with the white board perimeter

William has a special ability to use mathematics to predict special variables he can change to save someone's life

c.

I. If *Gupta Gupta* is standing at the ground 1.5 metres east of the origin , find the value(s) of p such that the beam will hit his feet (2 Marks)

II. Find the value(s) of p such that the area bounded by the graph of $f(x)$, the co-ordinate axes and the vertical line that *Gupta Gupta* stands on is equal to 389 m^2 , if this is not possible , **explain** why (3 Marks)



The incident was quite alarming, the school was sued for engineering flaws, but Mr. Kermond knew better: the matter was not of JKSS's structural integrity, rather of an external attack, and he had a pretty good idea of who it was



Question 2 (13 Marks)

TIME : 10:00 AM (GMT+10)

MEASUREMENT UNITS: Celcius

In order to capture the *Schmiddler* , Mr. Kermond designs a sophisticated booby trap. He attaches a special device that modulates the temperature of the *JKSS* building . The *rate* of the temperature change is modelled by

$$T(t) = \frac{50}{\pi} \cos(k(t - 3))$$

Where the temperature is measured in Celcius and t is the time in hours after 12:00 AM

The temperature cycles every 24 hours

a. Define the rule of $2T(\frac{t}{3})$

(1 Mark)

At 3:00 AM the temperature reaches $\frac{50}{\pi} + m$ Celcius , where m is the sum of integer values of k such the inequality $3 \sin(x) + 4 \cos(x) \leq k + 1$ has a solution

b.

i. Using calculus and algebra find m , (2 Marks)

ii. State the temperature at 2:00 PM

(3 Marks)

“But sir, how does this trap actually work?”

“Let me tell you William , we shall catch The *Schmiddler* once and for **all**”



. The *Schmiddler* has a strange medical condition, if the average rate of temperature is greater than g Celcius /hours , then he will fall ill

c.

I. Find the possible value(s) of g (2 Marks)

II. For what value(s) of g can the average rate of temperature change be equal to the average temperature? (2 Marks)

iii. Determine the value(s) of g that *schmiddler* will fall sick for atleast 1 hour **continuously** (2 Marks)

The workers at *Elementary Integrals* predicted Mr. Kermond's strategy, responding by turning on an AC at 8:50 AM , decreasing the temperature as governed by the function $h(t - a)$,where t is the time after 12:00 AM

d.

I. State the new temperature function $T_3(t)$ in terms of h (1 Mark)

ii . State the average temperature throughout the day if $h = 3$ (2 Marks)

William tries to break the AC to minimize h

iii. What's the minimum integer value of h so that if g is at it's minimum value, *Schmiddler* is still sick for atleast 1 hour (3 Marks)



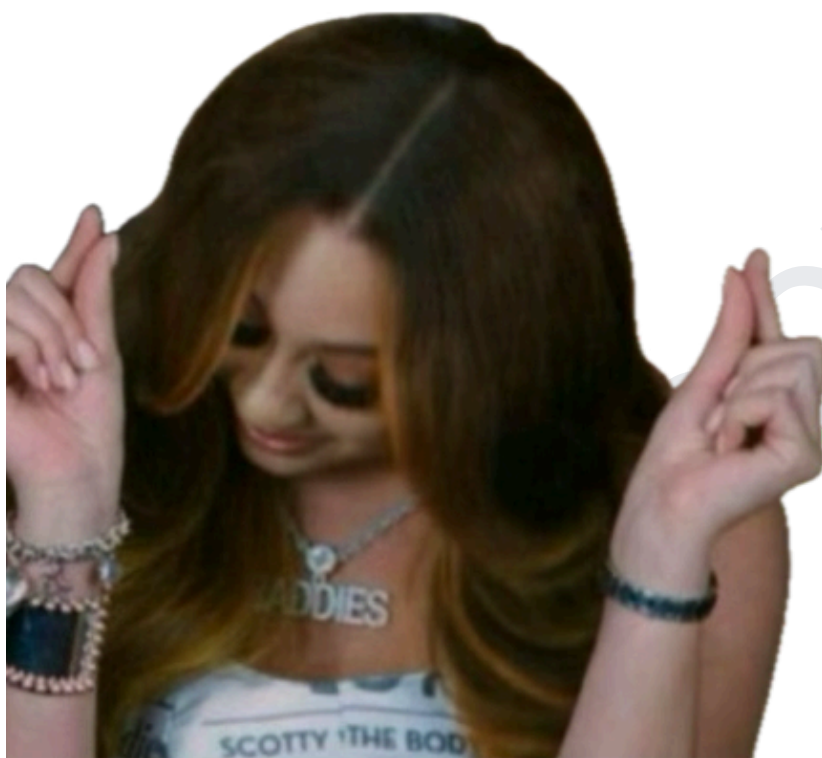
They manage to make the person sick but it turns out, - IT ISN'T ACTUALLY SCHMIDDLER, it was a decoy - a trick .

"DAMMIT"- Mr. kermond bellowed out to the hall

"He's tricked us!"

"Go to class William , we can't let the school know something is wrong"





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Question 3 (9 Marks)

TIME : 12:20 PM (GMT+10)

LOCATION: 2B3 (GOWS)

Agent Mann using the ML Parameters harnessed from the captured *BRIANLEAP-AI*'s database , has predicted that the *Schmiddler* will attack again at exactly 12:45 PM . Since William SAC-MAN) is most powerful whenever he does methods , *Agent Garrylescu* has rescheduled his class to this period, to **optimize** his chances of taking him down .

Will's substitute *Stephanie* has arranged the seats to form a peculiar shape, such that relative to a certain origin, the perimeter of the seats can be modelled by the function

$$f: [0, 6] \rightarrow \mathbb{R}, f(x) = 3e^x + 9$$

- a. Determine the points of intersection between $f(x)$ and $f^{-1}(x)$,if no such points exist, you must say so and explain why (2 Marks)

Stephanie moves the perimeter k units leftward

- b. State the value(s) of k for which $f(x)$ and $f^{-1}(x)$ have a non-zero bound area (2 Marks)

If the area bound by $f(x)$ and $f^{-1}(x)$ is **NOT** maximised, a pupil *Anu* will throw a tantrum in class

- c. Find the value(s) of k such that *Anu* will **not** throw a tantrum in class (2 Marks)



William believes the properties of the classroom encode Schmiddler's next attack

d.

I. If $k = 1$ Using algebra and **without** finding the inverse, determine the area bound by the graph of $f^{-1}(y)$ and the lines $y=3$ and $y=6$ (2 Marks)

II. Explain with the aid of appropriate graphs and diagrams what happens to the area bounded the chair perimeter and it's inverse as $k \rightarrow \infty$ (1 Mark)

Suddenly, William feels his perception disorienting as k increases without bound
His vision starts to blur, an EMP pulse bursts knocking out all power in JKSS



It was indeed a trap ...

If Schmidt was going down - he was taking down everyone with him

Men with guns barged in every corner of the room, but Kermond University could not be contacted as all alarm systems were disabled. He needed to get to the Panic room or it would've been too late

Question 4 (5 Marks)

TIME : 1:20 PM (GMT+10)

To enter *Mr. Kermond's* panic room , William must find it's area. *Mr. Kermond's* panic room has a perimeter , modelled by the function

$$p(x) = 3 \log_e(x - 3) + 9$$

Where x is relative a certain origin in 10s of metres while f is it's vertical height, in metres
90 metres east of the origin , a pillar stands to form a wall of the room
The room is uniformly 10 meters wide

- a. Using only **algebra** find the cross-sectional area of the panic room (3 Marks)

- b. Hence , state the Volume of the room (1 Mark)

- c. $p(x)$ is Dilated by a factor of k from the y - axis , such that the Volume is 1600 m^3 , state the value of k (1 Mark)

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After finding the area , William is cured of his dizziness

*“We Should be safe here , Will” - Mr. Kermond Stated as the school was placed in lockdown, Schmiddler’s **goons** were attacking from every direction , suddenly at the glint of a mirror, William spotted a hooded figure in the panic room, he thought it was a hallucination*

Or so he believed

Gupta Gupta Smiled in a creepy manner

Solve my problem , or die



Question 5 (7 Marks)

TIME : [DATA CORRUPTED] (GMT+X)

LOCATION: [DATE CORRUPTED]

MEASUREMENT:N/A

Gupta Gupta enjoys playing games, the only way for *Kermond* and William to stop him is by considering a mysterious function $f(x)$

It is known $\int_{24}^{60} f(x) dx = 9$

$$f(10) = 6$$

$$f(4) = 9$$

a.

I. Find the value of $\int_{24}^{60} 2f(x) + 3x dx$

1 Mark

II. Find the value of $\int_{-3}^3 f(6(x+7)) + x^2 dx$

(2 Marks)

William finds it with ease, *Gupta Gupta* realizes it will take more to break the mind of this warrior

b. Find the value of

(2 Marks)

$$\int_{-3/5}^3 f^{-1}(10x + 30) dy - \int_{-3/5}^3 f(20x + 60) dx$$



William is a bit more shaken , but is able to solve the problem despite it's monstrosity

"Im Tired of your games *Gupta* "

"STATE YOUR PURPOSE TO ME -STRAIGHT"

Gupta cackles again

"My behaviour is related to my own solicitude- thus the problems that permeate the recesses of my mind"

" $f(x)$ was always related to my mind, the average value of the parameter I speak of , is related to the function $g(x) = mf(n(x - a)) + c$ "

Suddenly a bullet whose paths follows the mysterious function attacks William, By finding it's average value from -1 to 1 he knows how to avoid the *expected* path

For part **b.**

Consider the case when $a = 0$

Assume m n and c are positive

c. Assume that over the interval $[-1,1]$ the Function has an average value of k

State all possible domains of which the average value of $g(x)$ from -1 to 1 is $km + c$
(2 Marks)

He wishes to find



It seems that Gupta Gupta's trap was just a paper tiger, the real lion was yet to come

Only The JKSS Chairs club has an answer for this madness

Question 6 (21 Marks)

TIME : 2:40 PM (GMT+X)

LOCATION: Chairs Club meeting at the underground lair

MEASUREMENT: Meters

To reach the Chairs Club Meeting he must repeat the steps he did for Mr. Kermond's panic room .

The security Camera's show *Schmiddle's* goons enclosing on the building, but the underground lair is hidden

The underground lair's entrance contains a river , whose northern boundary is modelled by

$$f_1: [0, 600] \rightarrow R, f_1(x) = 50 \sin\left(\frac{\pi x}{300}\right) + 200$$

$$f_2: [0, 600] \rightarrow R, f_2(x) = 50 \sin\left(\frac{\pi x}{300}\right) + k$$

Where x is the horizontal distance in metres and the graphs model the relation with height

The surface area of the river is believed to be 10,000 metres

a. State the value(s) of k

(1 Mark)

It turns out this is actually a lie by chairs club and $k = 150$

William wishes to swim from the southern boundary to the northern one, starting around 250 metres to the origin

b.

i. If William swims due north at 10 ms^{-1} , find the time taken to reach the northern boundary (1 Mark)



It turns out William forgot to bring his key card and must swim back
The key card scatters by a purely horizontal wind

ii. Determine the minimum possible time he can take to swim back (3 Marks)

Let p be the x-co-ordinate of the south boundary he swims back to, if he cannot get there in under 9 seconds, the card will de-activate

iii. Determine the value(s) of p such that the key card is not de-activated (2 Marks)

A security mechanism involves a laser whose line is modelled by
 $mx + 5$

Cuts through the boundaries of the river

c. Find the value(s) of m so that the area bounded by the laser the y axis and the river is bisected by the laser (3 Marks)



The Chairs Club executives wish to test him more , making him go through another room

He passes through a mirror whose perimeter is modelled (in the same dimensions as the river) by the family of equations :

$$g(x) = \sin(ax) + \cos(x), \text{ where } a \text{ is some integer}$$

d.

i. All local maxima of the mirror lie on the curve $g_1(x)$ and all local minima of the mirror lie on the curve $g_2(x)$ if $a \in R \setminus \{m\}$,state the rules of these curves , and the value of m giving a justification for your answer
(1 Mark)

ii. As a increases, the maximum value of $g(x)$ approaches a certain value, state this value, explaining why this occurs

William observes that the area bound by $g_1(x)$ and $g(x)$ from $x = 0$ and $x = 12\pi$ is equal to exactly $2\pi \text{ m}^2$

iii. State the possible value(s) of a (2 Marks)



But the game still isn't over yet and William must the final section before he get can get to *Schmittle* at the chairs club meeting room, he will do anything to keep him out

A sound wave begins to attack William and is initially modelled by the function

$$s(x) = (x^2 - 11x)\cos(x) + 9$$

Where s is the displacement of the wave from an equilibrium position and x is the distance from the room of the meeting

e. If William stands 5 meters east of the room, and the maximum range of the wave from his perception is 260 metres, then using only algebra, find all possible domain(s) of $s(x)$ (3 Marks)

Genuinely tired of these questions, he uses his powers to break open the chairs club meeting room by the transforming the wave to the function

$$10e^{-bt}\cos(2t) + 1$$

f. Determine the value(s) of b , such that **ratio** of the area bound by the curve tangent to all local maxima of the new wave and the wave itself, from the 6th maximum to 6th local minimum and the area bound by the wave and the curve tangent to all local minima of the wave from the 7th local maximum to the 7th local minimum is $e^{\frac{3\pi}{2}}$ (3 Marks)





[illegible]

Question 8 (6 Marks)

It turns out the chairs club meeting never happened, it was all an ethoscope of his reality
The Schmiddler left a note though , a clue of sorts:

“Consider the function $f(n) = n!$

Where $n! = n * (n - 1) * ... 1$

Use the fact that $(a + b)^n = \sum_{k=0}^n \frac{n!}{k!(n-k)!} a^{n-k} b^k$,

To approximate the value of π , correct to 3 decimal places (6 Marks)

SD-HBA-C



It appears that William Patel Never actually got to the Schmiddler, it was all a game in his mind, but his next entrance was revealed (What will it possibly be?!)

END OF QUESTION AND ANSWER BOOK

